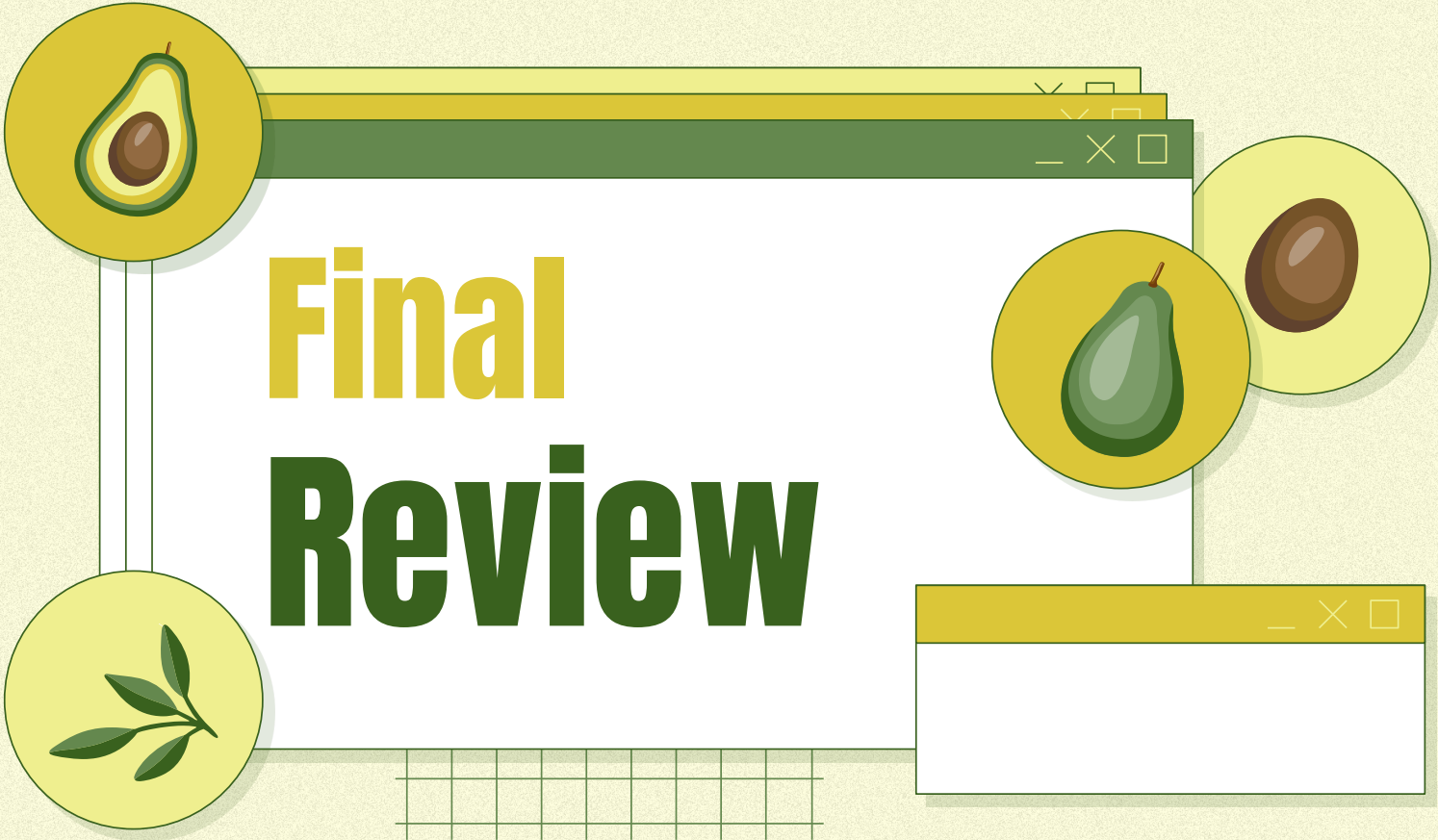




Final Review



Today's AGENDA

Final Logistics

Proctoring guidelines

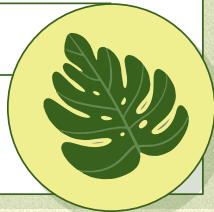
Formatting instructions

Question 1: Multiple Linear Regression and Feature Engineering

Question 2: Combinatorics

Question 3:

Question 4:



Exam Logistics

Proctoring
Formatting

Zoom link diff from mt → Piazza

Exam PDF timing

Midterm Logistics

Format

160 minutes, on paper, no calculators or electronics except viewing the exam and proctoring

15 minute submission window (you will have until 2:35pm PT)

Content

Lectures 1-9, Homeworks 1-8, Groupworks 1-4

When and Where

Friday, July 31st, 11:30am-2:30pm PT
Zoom

Scanning Test

- 24 pieces of paper → 48 pages
- Compiled into a PDF
- Share to computer
- Upload to gradescope
- Assign pages, one assignment per page of PDF
- 8:58:46

02

Winter 2024 Final Exam Part 1 Problem 4

Multiple Linear
Regression and Feature
Engineering

Albert collected 400 data points from a radiation detector. Each data point contains 3 features: feature A , feature B and feature C . The true particle energy E is also reported. Albert wants to design a linear regression algorithm to predict the energy E of each particle, given a combination of one or more of feature A , B , and C . As the first step, Albert calculated the correlation coefficients among A , B , C and E . He wrote it down in the following table, where each cell of the table represents the correlaton of two terms:

	A	B	C	E
A	1	-0.99	0.13	0.8
B	-0.99	1	0.25	-0.95
C	0.13	0.25	1	0.72
E	0.8	-0.95	0.72	1

C

Albert further refines his algorithm by fitting a prediction rule of the form:

$$H(A, B, C) = w_0 + w_1 \cdot A \cdot C + w_2 \cdot B^{C-7}$$

Given this prediction rule, what are the dimensions of the design matrix X ?

of data
pts

$$\downarrow \begin{bmatrix} \\ \\ \end{bmatrix}_{r \times c}$$

3 x 3

$$400 \times 3$$

So, what are r and c in r rows $\times$ c columns?

$$\begin{bmatrix} 1 & A_1 C_1 & B_1^{C_1-7} \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix} = \vec{w}$$

Spring 2024

01

Final Exam

Problem 8

Combinatorics

Now, suppose that each time you board a flight, a pilot hands you 1 card drawn at random from the set of possible cards. You fly 9 times, which gives you 9 cards drawn at random **with replacement** from the set of possible cards.

a Given that you receive a gold-colored card at least once, what is the probability that you receive a gold-colored card exactly 5 times? Show your work, and put a box around your final answer. $\rightarrow 0.0 + 0.9$

your final answer.

$P(\text{receive gold exactly } 5x)$

B

$\text{received a gold card } \geq 1x$

A

~~$P(A \cap B)$~~ $P(B|A) = \frac{P(A \cap B)}{P(A)}$

$\frac{P(\text{receive gold exactly } 5x \text{ out of } 9)}{P(\text{receive a gold card } \geq 1x)}$

$P(A \cap B) = P(B)$

$P(\text{Both exactly } 5 = \text{gold AND } \geq 1 \text{ is gold})$

w/ rep	ord matters	do not matter
VCSD PIDS		X
w/out rep	permutation	combinations

A Venn diagram with a large rectangle labeled 'S' (sample space). Inside 'S' is a circle labeled 'A'. Inside circle 'A' is a smaller circle labeled 'B'. This represents that event B is a subset of event A.

$$= \frac{P(\text{receive gold exactly } 5 \text{ out of } 9)}{1 - P(\text{never see a gold card})}$$

$$P(\text{receive gold exactly } 5 \text{ out of } 9) = \binom{9}{5} \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^4$$

$$P(\text{no gold cards}) = \left(\frac{2}{3}\right)^9$$

$$\frac{\frac{2}{3}}{1} \frac{\frac{2}{3}}{3} \frac{\frac{2}{3}}{3} \frac{\frac{2}{3}}{3} \frac{\frac{2}{3}}{3} \frac{\frac{2}{3}}{3} \frac{\frac{2}{3}}{3} \frac{\frac{2}{3}}{3} \frac{\frac{2}{3}}{9}$$

* at least
complement

$$\binom{9}{2} \binom{9}{3} = \binom{10}{3}$$

$$= \frac{\binom{9}{5} \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^4}{1 - \left(\frac{2}{3}\right)^9}$$

Delta has released a line of trading cards. Each trading card has a picture of a plane and a color. There are 12 plane types, and for each plane type there are three possible colors (gold, silver, and platinum), meaning there are 36 trading cards total.

Now, suppose that each time you board a flight, a pilot hands you 1 card drawn at random from the set of possible cards. You fly 9 times, which gives you 9 cards drawn at random **with replacement** from the set of possible cards.

In this question, you may leave your answers unsimplified, in terms of fractions, exponents, the permutation formula $P(n, k)$, and the binomial coefficient $\binom{n}{k}$.

b

The Airbus A350, the Boeing 767, and the Airbus A330 are 3 of the 12 plane types that Delta made trading cards for.

What is the probability that you receive 2 Airbus A350 cards, 4 Boeing 767 cards, and 3 Airbus A330 cards? Show your work, and put a box around your final answer.

$$2 + 4 + 3 = 9 \text{ out of } 9$$

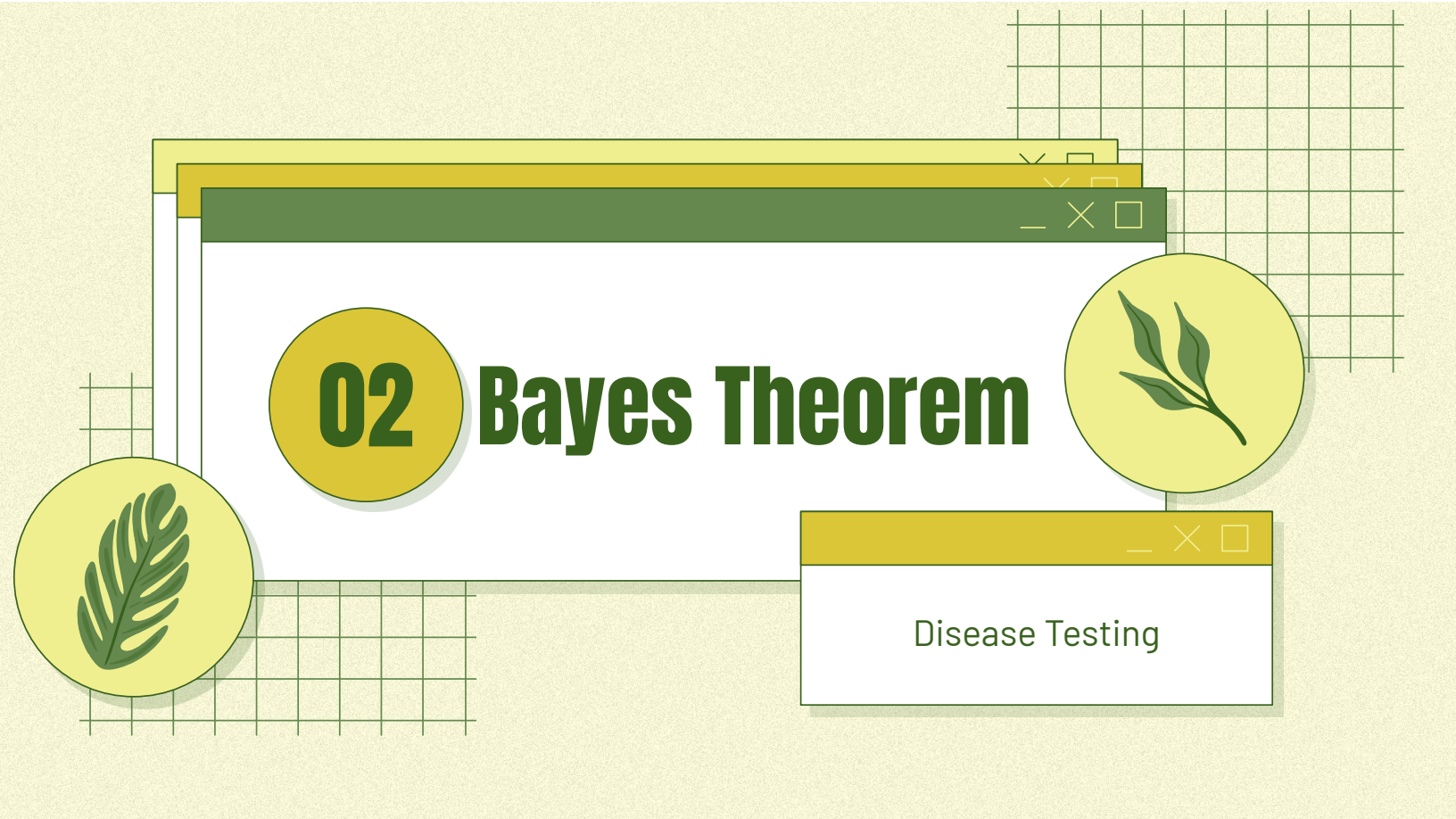
$$\binom{9}{2} \binom{7}{4} \binom{3}{3}$$

$$1 \Rightarrow 1$$

$$\left(\frac{1}{12}\right)^9 = \frac{1}{12} \frac{1}{12} \frac{1}{12} \frac{1}{12} \frac{1}{12} \frac{1}{12} \frac{1}{12} \frac{1}{12} \frac{1}{12}$$

$$\binom{9}{2} \binom{7}{4} \binom{3}{3} \cdot \frac{1}{12^9} = \frac{1}{12^9}$$

sequence



02 Bayes Theorem

Disease Testing

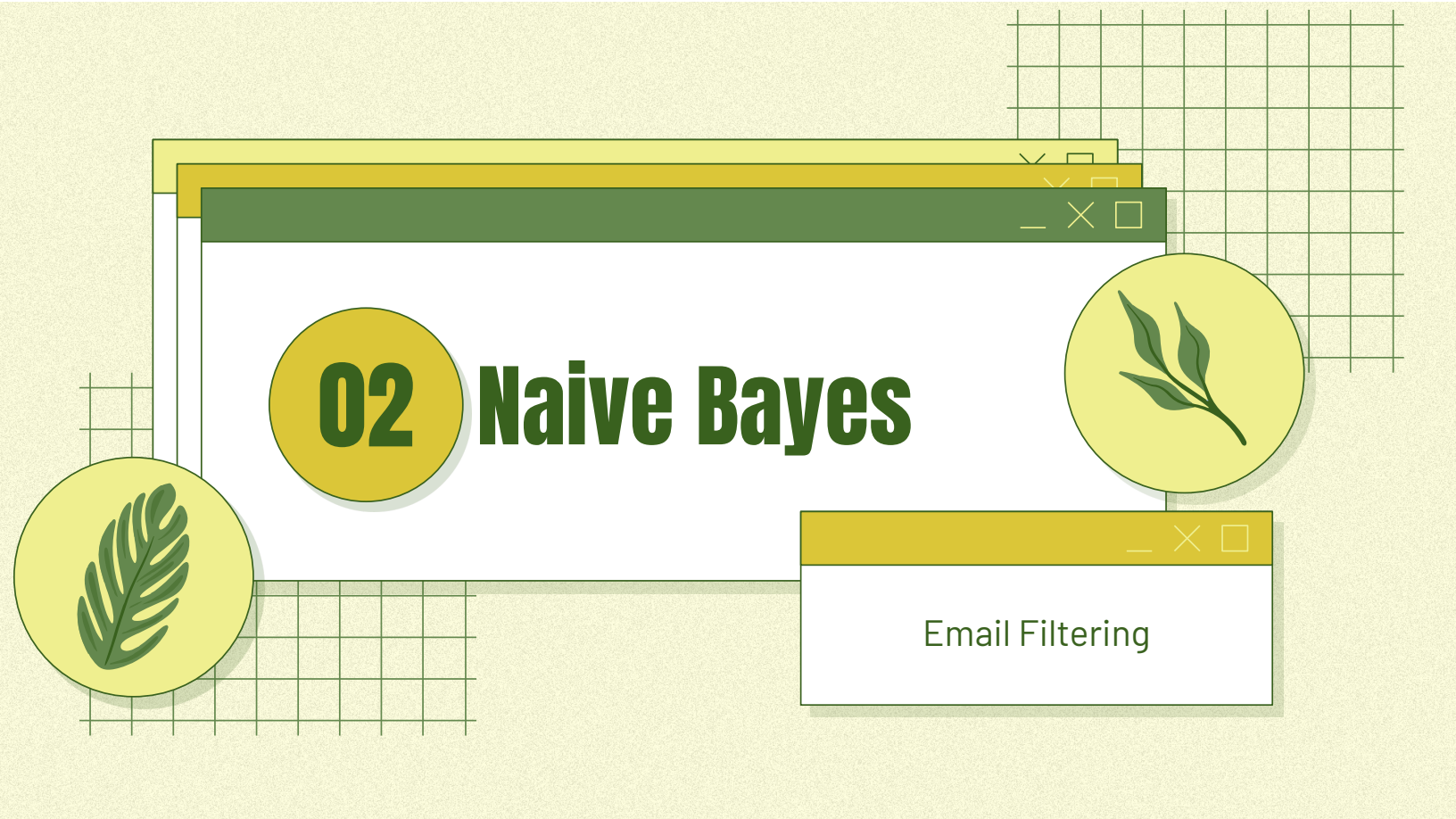
A rare disease affects just 1% of the population. Suppose D is the outcome that a given patient has the disease. The UCSD Hospital has developed a medical test with:

- Sensitivity (true positive rate): $P(+ \mid D) = 0.95$
- Specificity (true negative rate): $P(- \mid \overline{D}) = 0.90$

(a) A random person tests positive. Find $P(D \mid +)$.

(b) Suppose engineers improve the test's specificity to 0.99, with the sensitivity unchanged at 0.95. Recompute $P(D \mid +)$.

(c) Why does $P(D \mid +)$ jump so much in part (b), when sensitivity didn't change at all? What does this reveal about which error type matters most when the base rate is low?



02 Naïve Bayes

Email Filtering

A spam filter uses two features: whether a message contains the word "free" and whether it contains the word "meeting". From training data:

$$P(\text{Spam}) = 0.4, \quad P(\text{Ham}) = 0.6,$$

	Spam	Ham
$P(\text{"free"} \mid \cdot)$	0.8	0.1
$P(\text{"meeting"} \mid \cdot)$	0.05	0.4

(a) A new message contains "free" but not "meeting." Using Naive Bayes, compute the unnormalized scores for Spam and Ham and classify the message. Then give the normalized probability of Spam.



(b) Now suppose the message contains both "free" and "meeting." Classify it.

(c) Why is this method called "naive"? What independence assumption is it making about "free" and "meeting," and can you think of a realistic scenario where that assumption would be violated?

04

Advice and Recommendations

From Jordan and Eric

Advice

Breathe

Studies show that mindfulness can improve exam performance by freeing up brain space from test anxiety

Scavenger Hunt

The exam may or may not be in order of difficulty (which is subjective anyway), so start with what you know

Get Rest

Consistent sleep is an indicator of success.

Explainability

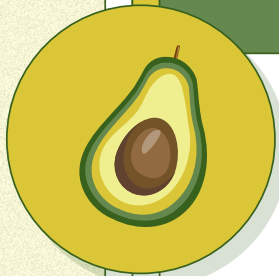
The easiest way to see that you've defended your proof properly is to imagine explaining it to someone with what you've written.

Don't be Hungry

Make sure to eat something before your exam. It'll give a good boost to your blood sugar and keep you energized. In fact, because you have a breakout room to yourself, keep a snack nearby!

Stay Hydrated

Keep a water bottle nearby! You never know when it'll come in handy.



THANKS!

DO YOU HAVE ANY QUESTIONS?

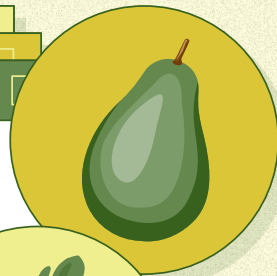
youremail@freepik.com

+91 620 421 838

yourwebsite.com



Please keep this slide for attribution



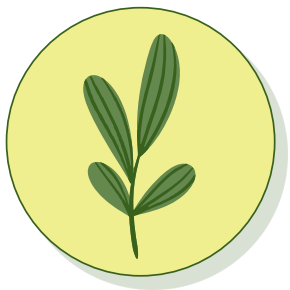
CREDITS: This presentation template
was created by **Slidesgo**, including
icons by **Flaticon** and infographics &
images by **Freepik**

ALTERNATIVE RESOURCES

Here's an assortment of alternative resources whose style fits the one of this template:

VECTORS:

- Flat summer pattern design
- Flat tropical summer background with leaves
- Flat summer pattern design (II)



RESOURCES

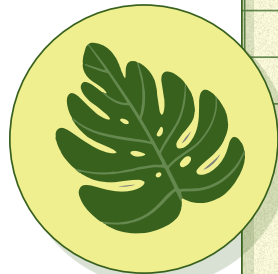
Did you like the resources on this template? Get them for free at our other websites:

PHOTOS:

- Top view delicious salad in an avocado
- Top view of avocado with asparagus and spinach
- Woman standing against wall drinking avocado smoothies in plastic cup
- African american man smiles

VECTORS:

- Flat summer pattern design
- Flat summer pattern design (II)
- Flat summer pattern design (III)
- Flat summer pattern design (IV)
- Flat tropical summer background with leaves



Instructions for use

If you have a free account, in order to use this template, you must credit [Slidesgo](#) by keeping the [Thanks](#) slide. Please refer to the next slide to read the instructions for premium users.

As a Free user, you are allowed to:

- Modify this template.
- Use it for both personal and commercial projects.

You are not allowed to:

- Sublicense, sell or rent any of Slidesgo Content (or a modified version of Slidesgo Content).
- Distribute Slidesgo Content unless it has been expressly authorized by Slidesgo.
- Include Slidesgo Content in an online or offline database or file.
- Offer Slidesgo templates (or modified versions of Slidesgo templates) for download.
- Acquire the copyright of Slidesgo Content.

For more information about editing slides, please read our FAQs or visit Slidesgo School:

<https://slidesgo.com/faqs> and <https://slidesgo.com/slidesgo-school>

Instructions for use (premium users)

As a Premium user, you can use this template without attributing [Slidesgo](#) or keeping the [Thanks](#) slide.

You are allowed to:

- Modify this template.
- Use it for both personal and commercial purposes.
- Hide or delete the "Thanks" slide and the mention to Slidesgo in the credits.
- Share this template in an editable format with people who are not part of your team.

You are not allowed to:

- Sublicense, sell or rent this Slidesgo Template (or a modified version of this Slidesgo Template).
- Distribute this Slidesgo Template (or a modified version of this Slidesgo Template) or include it in a database or in any other product or service that offers downloadable images, icons or presentations that may be subject to distribution or resale.
- Use any of the elements that are part of this Slidesgo Template in an isolated and separated way from this Template.
- Register any of the elements that are part of this template as a trademark or logo, or register it as a work in an intellectual property registry or similar.

For more information about editing slides, please read our FAQs or visit Slidesgo School:

<https://slidesgo.com/faqs> and <https://slidesgo.com/slidesgo-school>